# Supplementary Material for "Special optimal thermal cells: D-optimal thermal displacements for finite-temperature effective-harmonic phonons"

Albert Linda
*Department of Materials Science and Engineering, Indian Institute of Technology Kanpur, Kanpur 208016, India*

## S1. SURROGATE DESIGN STUDY

Table S1 lists the median phonon-mesh errors of six displacement designs on the DFT-fitted FC2+FC3+FC4 surrogates of Al, W, and NaCl. The surrogate effective FC2 was fitted to 6000 paired canonical samples; fitting half of them changes the mesh frequencies by 0.002 (Al), 0.001 (W), and 0.006 THz (NaCl). All designs were drawn from the harmonic FD model of the same $3 \times 3 \times 3$ cell and fitted in the space-group adapted FC2 basis with the cutoffs of the main text. Held-out forces on 200 canonical configurations follow the same ordering.

## S2. DESIGNS IN THE CRYSTAL-SYMMETRY BASIS

Table S2 repeats Table V of the main text with all fits in the FC2 basis of the full crystal space group (hiPhive cluster space, 3–12 parameters, cutoff limited to half the shortest image distance). For Si and Mg the two bases coincide.

Table S3 repeats the DFT sTDEP comparison of the main text in the crystal-symmetry basis.

## S3. REGRESSION FRAMEWORKS

Table S4 compares nine regression methods of the trainstation package, as used in hiPhive workflows, on the same DFT data and the same crystal-symmetry cluster space. Hyperparameters were selected by the package's internal cross-validation. All methods agree to within 0.03 THz.

## S4. DERIVATIONS FOR THE ESTIMATOR THEORY

*Effective FC2 as averaged Hessian.* For $\mathbf{u} \sim N(\mathbf{0}, \mathbf{C})$ and smooth $V$, Stein's lemma states $\langle \mathbf{u}\, g(\mathbf{u}) \rangle = \mathbf{C} \langle \nabla g \rangle$. With $g = \mathbf{F}^{\mathrm{T}} = -\nabla V^{\mathrm{T}}$ this gives $\langle \mathbf{u}\mathbf{F}^{\mathrm{T}} \rangle = -\mathbf{C} \langle \nabla\nabla V \rangle$, so the population least-squares coefficient of $\mathbf{F}$ on $\mathbf{u}$ is $-\langle \mathbf{F}\mathbf{u}^{\mathrm{T}} \rangle \mathbf{C}^{-1} = \langle \nabla\nabla V \rangle$. The SCHA free-energy gradient with respect to the auxiliary FC2 vanishes when $\langle \nabla\nabla V \rangle_{\mathbf{C}(\mathbf{\Phi})} = \mathbf{\Phi}$ for a fixed centroid, which is the self-consistent fixed point of the main text.

*Fixed-amplitude bias.* In mass-weighted normal coordinates with forces $f_a = -\omega_a^2 Q_a - \frac{1}{2} V_{abc} Q_b Q_c - \frac{1}{6} V_{abcd} Q_b Q_c Q_d$, the regression coefficient for a design with independent symmetric coordinates is $\Phi_{ae} = -\langle f_a Q_e \rangle / \langle Q_e^2 \rangle$. The cubic term vanishes by symmetry. The quartic term needs $\langle Q_b Q_c Q_d Q_e \rangle$, which for distinct indices paired as $(bb)(ee)$ equals $\sigma_b^2 \sigma_e^2$ for both Gaussian and $\pm\sigma$ coordinates, and for $b = c = d = e$ equals $3\sigma_e^4$ (Gaussian) or $\sigma_e^4$ (fixed amplitude). Collecting the three pairings,

$$\Phi_{ae} = \omega_a^2 \delta_{ae} + \tfrac{1}{2} \sum_b V_{abbe} \sigma_b^2 - \tfrac{1}{6} (3 - \kappa_e)\, V_{aeee} \sigma_e^2, \quad \text{(S1)}$$

with $\kappa_e = \langle Q_e^4 \rangle / \sigma_e^4$. The last term vanishes for Gaussian coordinates ($\kappa_e = 3$) and equals $-\frac{1}{3} V_{aeee} \sigma_e^2$ for the fixed-amplitude design ($\kappa_e = 1$). The statement is verified numerically by exact enumeration of all $2^6$ sign patterns of a random six-mode quartic model in the test suite (`test_optimal_design.py`), as are the exact cancellation of arbitrary cubic forces by $\pm$ pairs, the thermal mean-square displacements of the ZG and rattle designs, and, for a crystal with a two-atom basis, the canonical nearest-neighbor bond-stretch variance of the ZG configurations, which requires the Bloch phase of the full atomic position.

## S5. HIGHER-ORDER FORCE-CONSTANT VALIDATION

Figure S1 documents the joint FC2+FC3+FC4 fits used as surrogates: held-out errors of the nested models, convergence with the number of training snapshots, sensitivity to the interaction cutoffs, and comparison with the force changes produced by higher plane-wave cutoffs and denser $k$ meshes. For Al the denser-$k$-mesh force change exceeds the final fit error, so the Al tensors are used here only as model potentials for the sampling study, not as converged inputs for transport calculations.

TABLE S1. Surrogate design study (Al 300 K, W 1500 K, NaCl 700 K, Si 600 K; DFT-fitted FC2+FC3+FC4 surrogates in $3 \times 3 \times 3$ cells). Median mesh RMSE (THz) against the surrogate effective FC2 over 30 repetitions, at 2/4/8/16 force calls.

| Design | Al | W | NaCl | Si |
|---|---|---|---|---|
| Canonical | .199/.123/.088/.064 | .091/.068/.048/.033 | .224/.185/.116/.089 | .191/.122/.079/.064 |
| Canonical ± | .066/.054/.038/.032 | .035/.027/.021/.018 | .176/.127/.078/.078 | .060/.045/.033/.027 |
| Exact ampl., random signs | .066/.050/.038/.025 | .035/.028/.017/.016 | .166/.131/.093/.063 | .052/.042/.032/.018 |
| SOTC-D | .063/.056/.040/.025 | .039/.024/.021/.015 | .150/.131/.090/.062 | .054/.040/.031/.021 |
| Stratified ampl. + D-opt. | .073/.046/.038/.028 | .035/.031/.020/.015 | .151/.134/.088/.089 | .065/.039/.030/.023 |
| Gaussian ampl. + D-opt. | .074/.053/.046/.030 | .048/.032/.024/.019 | .184/.124/.114/.077 | .064/.043/.032/.025 |
| Special displ. (ZG) | .084/.067/.048/.041 | .042/.033/.017/.033 | .163/.133/.161/.174 | .036/.033/.022/.025 |
| Thermal rattle | .089/.074/.069/.076 | .053/.038/.032/.030 | .300/.295/.305/.323 | .313/.300/.296/.309 |
| Harmonic vs effective | 0.229 | 0.149 | 0.765 | 0.225 |

TABLE S2. Updated trial (iteration 1, crystal basis). Mesh RMSE (THz) against the iteration-1 effective FC2. Matched trial: SOTC-D designed from the iteration-1 trial ($D_1$) versus the better of random canonical subsets and pairs drawn from the same trial ($C_1$), scored against the disjoint effective FC2. Equal cost: bootstrapped SOTC-D (B; first pair from the harmonic trial, later pairs from the model refitted to the first pair, no extra force calls) and one-shot SOTC-D ($D_0$) versus canonical pairs from the harmonic trial ($C_0$; a single sTDEP iteration), scored against the 40-snapshot iteration-1 effective FC2, which shares no configuration with them. B and $D_0$ coincide at 2 calls by construction. The best entry of each group is bold.

| | matched, 2 | | equal cost, 2 | | | matched, 8 | | equal cost, 8 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $D_1$ | $C_1$ | B | $D_0$ | $C_0$ | $D_1$ | $C_1$ | B | $D_0$ | $C_0$ |
| Al | **0.06** | 0.08 | **0.07** | 0.07 | 0.10 | **0.03** | 0.05 | **0.03** | 0.05 | 0.04 |
| Ag | **0.04** | 0.11 | **0.02** | 0.02 | 0.13 | 0.06 | **0.05** | **0.06** | 0.09 | 0.10 |
| Au | 0.08 | **0.08** | **0.15** | 0.15 | 0.24 | **0.05** | 0.05 | **0.02** | 0.17 | 0.22 |
| Cu | **0.08** | 0.15 | 0.20 | 0.20 | **0.07** | **0.04** | 0.08 | 0.17 | **0.10** | 0.14 |
| Mo | 0.06 | **0.06** | **0.05** | 0.05 | 0.07 | **0.02** | 0.03 | **0.02** | 0.03 | 0.06 |
| W | 0.05 | **0.04** | 0.05 | 0.05 | **0.04** | **0.02** | 0.02 | 0.02 | **0.02** | 0.02 |
| NaCl | **0.05** | 0.11 | **0.11** | 0.11 | 0.14 | 0.09 | **0.07** | **0.05** | 0.15 | 0.12 |
| MgO | **0.22** | 0.32 | 0.42 | 0.42 | **0.34** | **0.09** | 0.15 | **0.11** | 0.32 | 0.26 |
| PbTe | **0.02** | 0.07 | 0.19 | 0.19 | **0.13** | **0.02** | 0.03 | **0.03** | 0.12 | 0.09 |
| Si | **0.07** | 0.10 | **0.10** | 0.10 | 0.11 | **0.05** | 0.05 | **0.05** | 0.09 | 0.07 |
| Mg | **0.05** | 0.06 | **0.06** | 0.06 | 0.07 | **0.03** | 0.05 | **0.02** | 0.06 | 0.04 |
| $D_1 < C_1$ / $B < C_0$ | 8/11 | | 7/11 | | | 9/11 | | 9/11 | | |
| held-out force $D_1 < C_1$ | 9/11 | | | | | 8/11 | | | | |

TABLE S3. Self-consistent protocols with DFT forces, equal cost. Mesh RMSE (THz) against the 40-snapshot iteration-1 effective FC2 (crystal basis). B: bootstrapped SOTC-D. sTDEP: half of the calls as canonical ± pairs from the harmonic trial, half as canonical ± pairs from the model refitted to them, fitted to the second iteration only (last) or to both (pool); median of three seeds. $C_0$: one-shot canonical pairs from the harmonic trial (median of 20 subsets). No configuration of any protocol enters the reference. The best entry of each group is bold.

| | 8 calls | | | | 16 calls | | | |
|---|---|---|---|---|---|---|---|---|
| | B | sTDEP last | sTDEP pool | $C_0$ | B | sTDEP last | sTDEP pool | $C_0$ |
| Al | **0.03** | 0.05 | 0.05 | 0.04 | 0.02 | 0.03 | **0.01** | 0.03 |
| Ag | 0.06 | **0.04** | 0.07 | 0.10 | 0.04 | **0.03** | 0.09 | 0.10 |
| Au | **0.02** | 0.06 | 0.17 | 0.22 | **0.01** | 0.04 | 0.15 | 0.21 |
| Cu | 0.17 | 0.18 | **0.10** | 0.14 | 0.16 | 0.08 | **0.06** | 0.14 |
| Mo | **0.02** | 0.05 | 0.03 | 0.06 | 0.02 | 0.03 | **0.02** | 0.05 |
| W | 0.02 | 0.05 | 0.03 | **0.02** | 0.02 | 0.03 | **0.02** | 0.02 |
| NaCl | **0.05** | 0.13 | 0.09 | 0.12 | **0.04** | 0.10 | 0.13 | 0.12 |
| MgO | **0.11** | 0.22 | 0.18 | 0.26 | **0.10** | 0.19 | 0.22 | 0.25 |
| PbTe | **0.03** | 0.05 | 0.07 | 0.09 | **0.04** | 0.04 | 0.06 | 0.08 |
| Si | **0.05** | 0.08 | 0.08 | 0.07 | **0.04** | 0.05 | 0.06 | 0.06 |
| Mg | **0.02** | 0.05 | 0.03 | 0.04 | **0.02** | 0.04 | 0.04 | 0.03 |
| B best | | 8/11 | | | | 6/11 | | |

TABLE S4. Regression frameworks (hiPhive cluster space with crystal symmetry, trainstation fit methods) on the same DFT data, 8 force calls. Entries: mesh RMSE (THz) against the disjoint effective FC2 (supercell basis, so the entries include the basis difference listed in the main text) for random canonical subsets (median of 10) / SOTC-D.

| Method | Al | Ag | Au | Cu | Mo | W | NaCl | MgO | PbTe | Si | Mg |
|---|---|---|---|---|---|---|---|---|---|---|---|
| least-squares | 0.20/0.18 | 0.11/0.09 | 0.12/0.13 | 0.23/0.20 | 0.09/0.08 | 0.06/0.05 | 0.13/0.11 | 0.31/0.26 | 0.06/0.11 | 0.10/0.05 | 0.08/0.04 |
| ridge | 0.20/0.18 | 0.11/0.09 | 0.12/0.13 | 0.24/0.20 | 0.09/0.08 | 0.06/0.05 | 0.13/0.12 | 0.31/0.26 | 0.06/0.12 | 0.10/0.05 | 0.08/0.04 |
| bayesian-ridge | 0.20/0.18 | 0.11/0.09 | 0.12/0.14 | 0.24/0.20 | 0.09/0.08 | 0.06/0.05 | 0.13/0.12 | 0.31/0.26 | 0.06/0.12 | 0.10/0.05 | 0.08/0.03 |
| ardr | 0.20/0.18 | 0.11/0.09 | 0.12/0.14 | 0.24/0.20 | 0.09/0.08 | 0.06/0.05 | 0.14/0.11 | 0.32/0.27 | 0.07/0.12 | 0.08/0.06 | 0.07/0.04 |
| rfe | 0.20/0.18 | 0.11/0.09 | 0.12/0.13 | 0.23/0.20 | 0.09/0.08 | 0.06/0.05 | 0.13/0.11 | 0.33/0.26 | 0.06/0.12 | 0.10/0.07 | 0.07/0.04 |
| omp | 0.20/0.18 | 0.11/0.09 | 0.12/0.13 | 0.23/0.20 | 0.09/0.08 | 0.06/0.05 | 0.14/0.11 | 0.31/0.26 | 0.06/0.11 | 0.10/0.08 | 0.07/0.04 |
| lasso | 0.20/0.18 | 0.11/0.09 | 0.12/0.13 | 0.23/0.20 | 0.09/0.08 | 0.06/0.05 | 0.14/0.11 | 0.31/0.26 | 0.06/0.11 | 0.09/0.06 | 0.07/0.04 |
| adaptive-lasso | 0.20/0.18 | 0.13/0.12 | 0.12/0.16 | 0.23/0.20 | 0.09/0.08 | 0.06/0.05 | 0.14/0.11 | 0.31/0.26 | 0.06/0.11 | 0.09/0.05 | 0.08/0.06 |
| elasticnet | 0.20/0.18 | 0.11/0.09 | 0.12/0.13 | 0.23/0.20 | 0.09/0.08 | 0.06/0.05 | 0.14/0.11 | 0.31/0.26 | 0.06/0.11 | 0.09/0.06 | 0.07/0.04 |

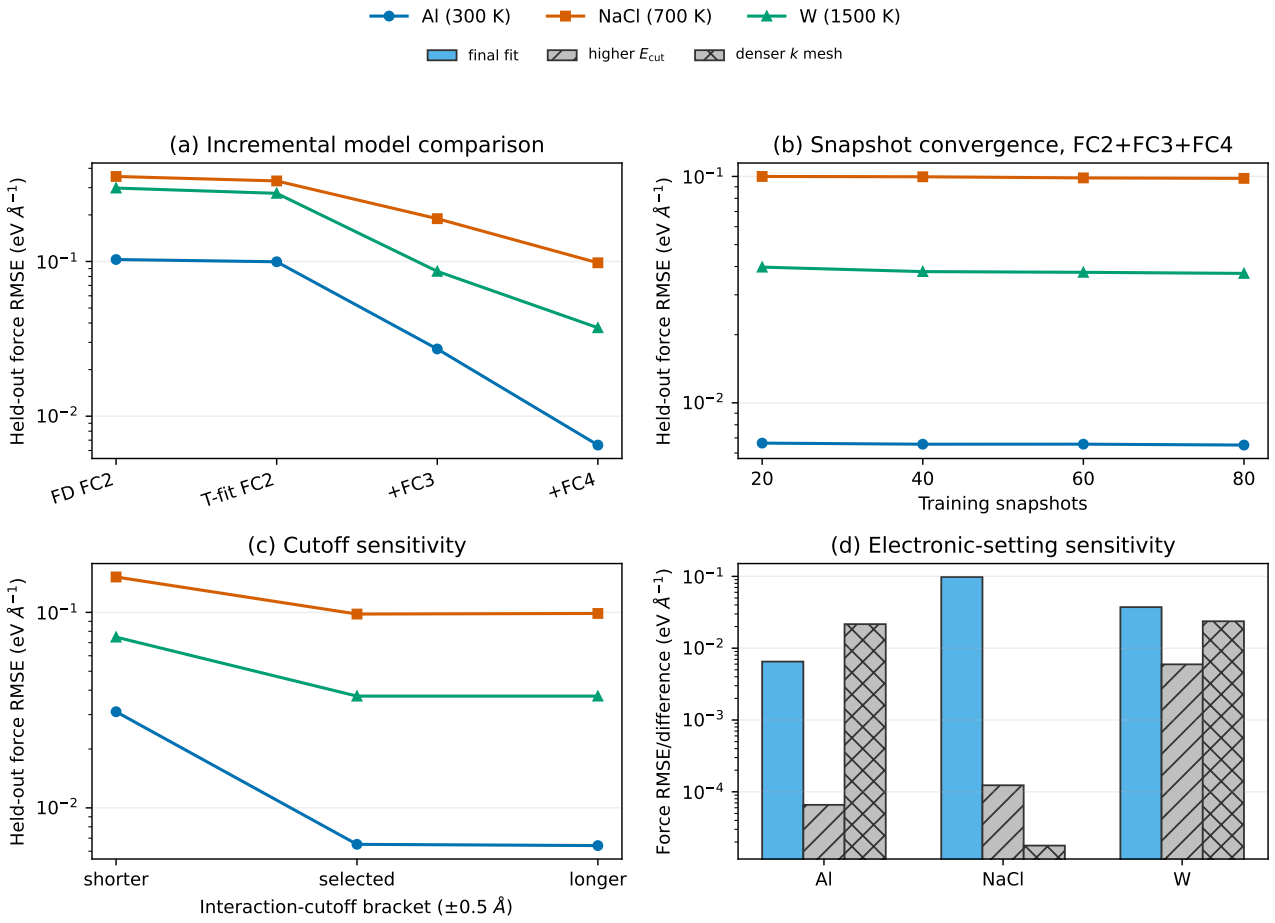



FIG. S1. Validation of the FC2+FC3+FC4 fits to 80 canonical QE snapshots (Al 300 K, W 1500 K, NaCl 700 K; $3 \times 3 \times 3$ cells). (a) Held-out force RMSE of harmonic FD FC2 and of temperature-fitted FC2, FC2+FC3, and FC2+FC3+FC4 models. (b) Convergence with training snapshots. (c) Cutoff sensitivity (±0.5 Å). (d) Fit error compared with electronic-setting force differences.

## S6. INDIVIDUAL ITERATIONS OF THE SOTC-C RUNS

TABLE S5: Individual iterations of the SOTC-C runs: configurations, rms displacement, unstable mesh fraction and minimum frequency of the axial-prior fit, and normalized condition number of the space-group adapted design.

| Material | iter. | $N_{\mathrm{ens}}$ | $u_{\mathrm{rms}}$ (Å) | $f_{\mathrm{unst}}$ | $\omega_{\mathrm{min}}$ (THz) | cond. |
|---|---|---|---|---|---|---|
| Al | 1 | 5 | 0.106 | 0.000 | 0.60 | 5.2 |
| Al | 2* | 5 | 0.106 | 0.000 | 0.48 | 3.4 |
| Ag | 1 | 5 | 0.167 | 0.000 | 0.26 | 5.7 |
| Ag | 2 | 5 | 0.167 | 0.000 | 0.42 | 3.2 |
| Ag | 3 | 5 | 0.167 | 0.000 | 0.44 | 3.2 |
| Ag | 4* | 5 | 0.167 | 0.000 | 0.32 | 2.8 |
| Au | 1* | 1 | 0.186 | 0.007 | -0.95 | 10.7 |
| Au | 2 | 1 | 0.186 | 0.159 | -4.58 | 7.5 |
| Cu | 1 | 8 | 0.169 | 0.000 | 0.51 | 15.3 |
| Cu | 2 | 8 | 0.169 | 0.000 | 0.45 | 7.9 |
| Cu | 3* | 8 | 0.169 | 0.000 | 0.45 | 7.5 |
| Cu | 4 | 8 | 0.169 | 0.000 | 0.52 | 8.2 |
| Mo | 1 | 1 | 0.111 | 0.000 | 0.52 | 17.1 |
| Mo | 2* | 1 | 0.111 | 0.000 | 0.99 | 8.0 |
| W | 1 | 3 | 0.115 | 0.000 | 0.76 | 4.5 |
| W | 2* | 3 | 0.115 | 0.000 | 0.73 | 2.9 |
| W | 3 | 3 | 0.115 | 0.000 | 0.75 | 3.5 |
| W | 4 | 3 | 0.115 | 0.000 | 0.74 | 3.3 |
| NaCl | 1 | 3 | 0.191 | 0.000 | 0.31 | 3.9 |
| NaCl | 2 | 3 | 0.191 | 0.000 | 0.34 | 2.8 |
| NaCl | 3 | 3 | 0.191 | 0.000 | 0.32 | 2.9 |
| NaCl | 4 | 3 | 0.191 | 0.000 | 0.39 | 3.1 |
| NaCl | 5* | 3 | 0.191 | 0.000 | 0.31 | 2.4 |
| MgO | 1 | 3 | 0.141 | 0.000 | 0.96 | 2.9 |
| MgO | 2 | 3 | 0.141 | 0.000 | 0.73 | 2.1 |
| MgO | 3 | 3 | 0.141 | 0.000 | 0.96 | 2.6 |
| MgO | 4* | 3 | 0.141 | 0.000 | 0.88 | 2.2 |
| PbTe | 1 | 5 | 0.190 | 0.000 | 0.21 | 3.1 |
| PbTe | 2* | 5 | 0.190 | 0.000 | 0.22 | 2.2 |
| PbTe | 3 | 5 | 0.190 | 0.000 | 0.19 | 2.1 |
| PbTe | 4 | 5 | 0.190 | 0.000 | 0.22 | 2.4 |

An asterisk marks the iteration selected by the workflow. The two Au iterations each contain a single configuration; their axial-prior fits are unstable, whereas the pooled fit is stable.

## S7. LATTICE-PARAMETER SENSITIVITY

## S8. ACCURACY OVERVIEW

## S9. NACL OPTIC MODES AND NEUTRON DATA

## S10. THERMAL DISPLACEMENT CORRELATOR

In second quantization, the displacement of basis atom $a$ in cell $\mathbf{R}$ is

$$\hat{u}_a^{\alpha}(\mathbf{R}) = \frac{1}{\sqrt{N_c}} \sum_{\mathbf{q}s} \sqrt{\frac{\hbar}{2M_a\omega_{\mathbf{q}s}}}\, e_a^{\alpha}(\mathbf{q}s) \left(\hat{a}_{\mathbf{q}s} + \hat{a}^{\dagger}_{-\mathbf{q}s}\right) e^{i\mathbf{q}\cdot\mathbf{R}}. \tag{S2}$$

With $\langle \hat{a}^{\dagger}_{\mathbf{q}s}\hat{a}_{\mathbf{q}'s'}\rangle = n_B\delta\delta$, $\langle \hat{a}_{\mathbf{q}s}\hat{a}^{\dagger}_{\mathbf{q}'s'}\rangle = (n_B+1)\delta\delta$, vanishing anomalous averages, and $2n_B+1 = \coth(\hbar\omega/2k_BT)$, the canonical average $\langle \hat{u}_a^{\alpha}(\mathbf{0})\hat{u}_b^{\beta}(\mathbf{R})\rangle$ reduces to Eq. (4) of the main text: only $\mathbf{q}' = -\mathbf{q}$ contributes, and $e_b^{\beta}(-\mathbf{q}s) = e_b^{\beta *}(\mathbf{q}s)$ gives the factor $e_a^{\alpha}(\mathbf{q}s)e_b^{\beta *}(\mathbf{q}s)e^{-i\mathbf{q}\cdot\mathbf{R}}$. Its trace with the completeness relation of the polarization vectors gives $U_{\mathrm{iso}} = (\hbar/6N_cM)\sum_{\mathbf{q}s}\omega_{\mathbf{q}s}^{-1}\coth(\hbar\omega_{\mathbf{q}s}/2k_BT)$ for a monatomic crystal.

In the commensurate supercell the same covariance is diagonal in the normal-mode coordinates $Q_m$ with variances $\sigma_m^2$ given in Sec. II D of the main text. A configuration of the form of Eq. (7) has $Q_m^2 = \sigma_m^2$ for every mode, so that its single-configuration second moment equals the canonical one mode by mode; cross terms $Q_mQ_{m'} = s_ms_{m'}\sigma_m\sigma_{m'}$ average to zero only over the sign patterns, and their effect on the force fit is what the D-optimal sign choice controls.

## S11. SOTC OBJECTIVE GRADIENT

For the shell objective of Eq. (6) of the main text, write the cell correlator as a sum over ordered atom pairs, $\bar{C}^{\mathrm{cell}}_{2,ab}(R;\mathbf{u}) = \sum_{i,j} P_{ij}^{abR}\,\mathbf{u}_i\cdot\mathbf{u}_j$, where the symmetric weight matrix $P_{ij}^{abR}$ equals $1/(3N_R^{ab}N_a)$ (or half of it for each order when $a \neq b$) if the pair belongs to shell $R$ of the species channel and zero otherwise; the on-site term is the shell $R = 0$. Then

$$\frac{\partial \mathcal{Q}}{\partial \mathbf{u}_i} = \sum_{a,b,R} 4\, w_{ab}(R)\, \Delta_{ab}(R) \sum_j P_{ij}^{abR}\,\mathbf{u}_j, \tag{S3}$$

with $\Delta_{ab}(R) = \bar{C}^{\mathrm{cell}}_{2,ab} - \bar{C}_{2,ab}$, and the mode-amplitude gradient follows by the chain rule through $\mathbf{u}_i = \sum_m A_m\mathbf{e}_{im}/\sqrt{M_i}$. The optimization is carried out in mode amplitudes and phases with random restarts.

TABLE S6. PBE equilibrium lattice parameters from Birch–Murnaghan fits (primitive cells) and the effect of using them instead of the production values on the harmonic $3 \times 3 \times 3$ FD model: mesh RMSE and change in $S_{\mathrm{vib}}(T_{\mathrm{des}})$.

| | $a_{\mathrm{prod}}$ (Å) | $a_{\mathrm{PBE}}$ (Å) | $\Delta a/a$ (%) | $B_0$ (GPa) | RMSE (THz) | $\Delta S_{\mathrm{vib}}$ |
|---|---|---|---|---|---|---|
| Al | 4.050 | 4.039 | -0.26 | 78 | 0.10 | -0.38 |
| Ag | 4.120 | 4.146 | +0.63 | 91 | 0.16 | +1.17 |
| Au | 4.130 | 4.150 | +0.49 | 142 | 0.12 | +1.17 |
| Cu | 3.670 | 3.631 | -1.06 | 142 | 0.35 | -1.71 |
| Mo | 3.160 | 3.160 | -0.01 | 259 | 0.00 | -0.01 |
| W | 3.180 | 3.184 | +0.14 | 305 | 0.03 | +0.18 |
| NaCl | 5.730 | 5.704 | -0.45 | 24 | 0.12 | -0.64 |
| MgO | 4.260 | 4.249 | -0.25 | 153 | 0.14 | -0.30 |
| PbTe | 6.510 | 6.567 | +0.88 | 39 | 0.14 | +1.72 |

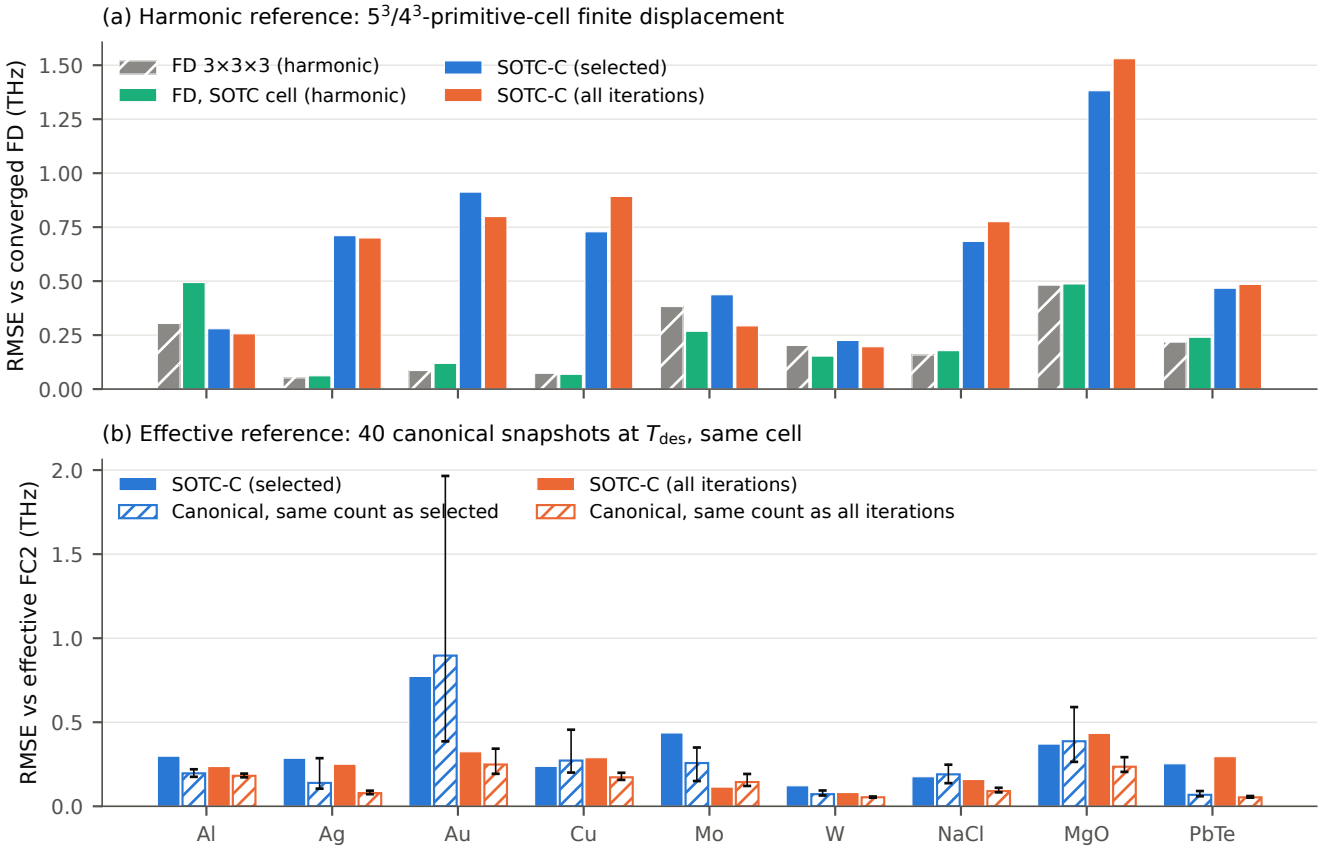



FIG. S2. (a) Mesh RMSE against the converged large-cell harmonic FD reference for harmonic FD in $3 \times 3 \times 3$ cells and in the SOTC cells, and for the selected and pooled SOTC-C models. (b) Mesh RMSE against the same-cell effective FC2 for the selected and pooled SOTC-C models and for random canonical subsets of the same size fitted with the same prior (median and 10–90% range over 20 draws).

## S12. BRANCH-RESOLVED CONVERGENCE AND NEUTRON MODEL FOR AL

Figure S4 shows the dispersions of SOTC-D from the updated trial with 2, 4, 8, and 16 force calls for Al (300 K) and Au (1200 K), together with the converged large-cell harmonic FD reference and the 40-snapshot iteration-1 effective FC2 of the same cell. For Au, whose single-configuration SOTC-C fits are unstable (Table S5), the SOTC-D dispersions are stable at every budget and approach the effective FC2 branch by branch; with two calls the largest deviations are in the transverse branches along

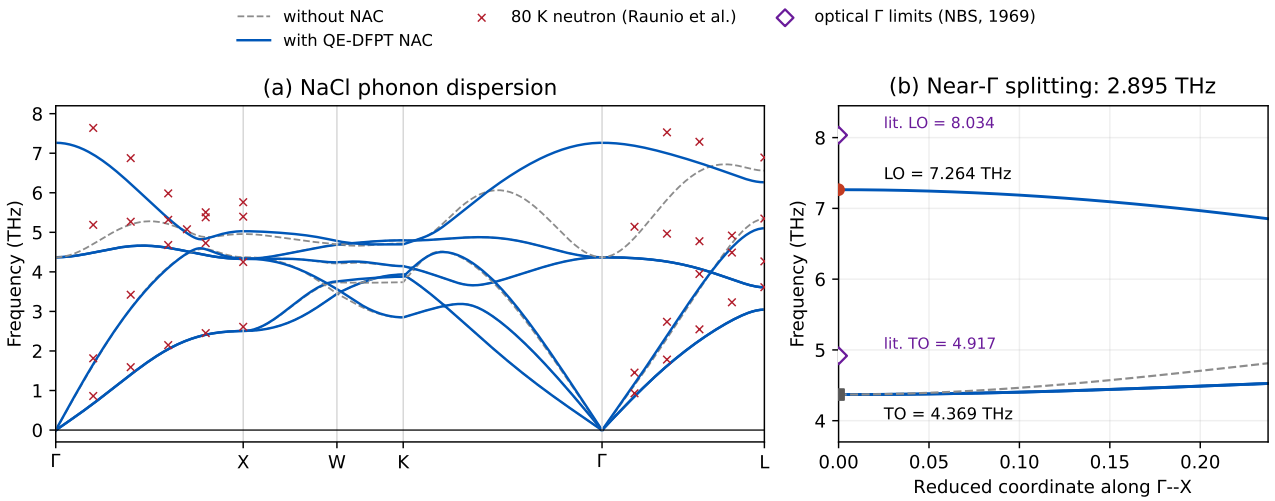



FIG. S3. NaCl harmonic FD FC2 ($3 \times 3 \times 3$ cell) without (gray dashed) and with (blue) the non-analytic correction from QE-DFPT Born charges and dielectric tensor. Red crosses: 39 neutron points at 80 K along Γ–X and Γ–L[1]; purple diamonds: infrared TO and LO limits[2]. The neutron data are temperature-mismatched to the fixed-volume calculation and are shown without a global error metric.

X–W–K and Γ–L. For Al the Born–von Kármán models fitted to the inelastic neutron spectrum at 10 and 300 K[3] (eight shells) are also shown. The converged harmonic reference agrees with the 10 K model within 1.1% at X and L. The 300 K model is softer in the transverse branches (TA(X) 5.39 THz) than both the converged harmonic reference (5.72 THz) and the effective FC2 of the 32-atom cell (6.01 and 6.29 THz, the degeneracy being lifted by the cell), most of whose stiffening is already present in the harmonic FD model of that cell (5.87 and 6.09 THz). A quantitative test of the finite-temperature renormalization against experiment is therefore limited by the supercell truncation of the 32-atom cell and by the non-uniqueness of density-of-states-fitted force-constant models for individual branches, especially away from the high-symmetry points.

## S13. DESIGNS AT FOUR AND SIXTEEN FORCE CALLS

Tables S7, S8, and S9 extend Tables IV and V of the main text and Table S2 to four and sixteen force calls, with the same definitions.

## S14. NEAREST-NEIGHBOR FORCE CONSTANTS AGAINST NEUTRON DATA

Table S10 compares the leading real-space force constants with Born–von Kármán fits to inelastic neutron data[4] where the measurement temperature matches the model: Al at 300 K and Cu at 1200 K (interpolated between 973 and 1336 K). For the other crystals the neutron fits are at room temperature, whereas the models are at 1000–1500 K and the corresponding lattice parameters, so they are not compared. The converged harmonic reference of Al reproduces the 300 K constants within 4%. In the 32-atom Al cell, truncation distorts individual constants (the second-shell constant falls to 0.5 N/m). The effective and SOTC-D models partly compensate, so shell constants of the small cell should not be read as converged. For Cu, where truncation is small ($K_1$ 23.5 N/m in both cells), the harmonic model at the 1200 K lattice parameter is 8% softer than experiment in $K_1$. The effective and SOTC-D models are 16–18% stiffer (bootstrapped SOTC-D 30%). The SCHA auxiliary FC2 omits the cubic (bubble) contribution to the physical spectrum, which softens it, so a stiffer auxiliary FC2 than the measured spectrum is the expected sign.

## S15. THERMODYNAMICS OF AL ON THE VOLUME GRID

Table S11 complements Table XIII of the main text. The Grüneisen formula $\alpha = \sum_k \gamma_k c_k / 3 B_T V$, with mode Grüneisen parameters from the converged harmonic FC2 on the volume grid, reproduces the free-energy minimization to $0.1 \times 10^{-6}$ K$^{-1}$ at every temperature when it is evaluated at $V(T)$ with $B_T$, which checks the numerics of the minimization. Evaluated with the static $B_0$ and $V_0$ (the common shortcut), it is 8% low at 100 K and 13–44% low at 300–900 K, because it omits the softening of $B_T$ and the volume dependence of the mode Grüneisen parameters; the shortcut is therefore not a valid check of a QHA calculation above about 100 K for Al. The converged QHA gives $C_P$ within 0.1–1.0 J mol$^{-1}$K$^{-1}$ of the NIST-JANAF values between 300 and 900 K,

TABLE S7. Matched comparison in the SOTC cell at equal numbers of displaced force calls, all designed from the same-cell harmonic FD trial and fitted in the supercell space-group basis. D: SOTC-D; C: random canonical subsets; P: random canonical $\pm\mathbf{u}$ pairs (medians over 20 draws); Z: special displacements with Walsh sign patterns (ZG); R: thermal rattle (independent Gaussian atoms with the canonical per-atom variance). Upper block: held-out force RMSE on 8 canonical test snapshots relative to that of the 40-snapshot effective FC2. Lower block: mesh RMSE (THz) against an effective FC2 fitted to $40 - n$ canonical snapshots disjoint from the scored design. The best entry is bold; the last row counts the materials in which D is best. Entries >10 THz (>100 in force) are singular fits of designs that do not identify the model.

| | 4 calls | | | | | 16 calls | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | D | C | P | Z | R | D | C | P | Z | R |
| Held-out force RMSE / floor | | | | | | | | | | |
| Al | **1.005** | 1.053 | 1.015 | 1.025 | 1.050 | **0.997** | 1.007 | 1.001 | 1.020 | 1.019 |
| Ag | **1.000** | 1.061 | 1.029 | 1.032 | 1.107 | **0.997** | 1.011 | 1.009 | 1.016 | 1.060 |
| Au | **1.004** | 1.048 | 1.024 | 1.034 | 1.130 | **1.002** | 1.010 | 1.005 | 1.020 | 1.110 |
| Cu | **1.038** | 1.154 | 1.186 | 1.567 | 1.366 | **1.006** | 1.032 | 1.013 | 1.026 | 1.073 |
| Mo | **1.006** | 1.071 | 1.023 | 1.021 | 1.040 | **0.999** | 1.011 | 1.003 | 1.018 | 1.010 |
| W | **1.007** | 1.057 | 1.017 | 1.026 | 1.024 | **0.997** | 1.010 | 1.002 | 1.010 | 1.005 |
| NaCl | 1.005 | 1.005 | 1.005 | 1.002 | **0.996** | 1.002 | 1.001 | 1.000 | 0.999 | **0.982** |
| MgO | **1.012** | 1.035 | 1.031 | 1.030 | 1.135 | **0.998** | 1.007 | 1.005 | 1.005 | 1.058 |
| PbTe | 1.027 | 1.018 | 1.018 | **1.011** | 1.045 | 1.012 | 1.003 | **1.001** | 1.016 | 1.141 |
| Si | **0.999** | 1.005 | 1.002 | 1.001 | 1.131 | **0.998** | 1.002 | 1.000 | 1.000 | 1.116 |
| Mg | **1.000** | 1.007 | 1.002 | 1.002 | 1.023 | 1.000 | 1.001 | 1.000 | **1.000** | 1.016 |
| D best | | | 9/11 | | | | | 8/11 | | |
| Mesh RMSE vs disjoint effective FC2 (THz) | | | | | | | | | | |
| Al | **0.11** | 0.25 | 0.14 | 0.15 | 0.27 | **0.08** | 0.14 | 0.10 | 0.12 | 0.19 |
| Ag | **0.11** | 0.32 | 0.21 | 0.24 | 0.41 | **0.10** | 0.18 | 0.14 | 0.18 | 0.30 |
| Au | **0.17** | 0.27 | 0.20 | 0.30 | 0.40 | **0.11** | 0.17 | 0.13 | 0.19 | 0.42 |
| Cu | **0.40** | 0.78 | 0.92 | 2.68 | 1.78 | 0.29 | 0.36 | **0.28** | 0.38 | 0.51 |
| Mo | **0.08** | 0.17 | 0.10 | 0.11 | 0.14 | 0.06 | 0.11 | **0.05** | 0.11 | 0.10 |
| W | **0.06** | 0.13 | 0.07 | 0.09 | 0.08 | **0.04** | 0.07 | 0.04 | 0.06 | 0.06 |
| NaCl | 0.25 | 0.29 | **0.24** | 0.34 | 0.59 | 0.15 | 0.17 | **0.14** | 0.15 | 0.66 |
| MgO | **0.40** | 0.57 | 0.52 | 0.47 | 1.05 | 0.34 | 0.39 | **0.30** | 0.36 | 0.74 |
| PbTe | 0.16 | 0.13 | 0.13 | **0.12** | 0.27 | 0.10 | **0.08** | 0.08 | 0.09 | 0.37 |
| Si | 0.04 | 0.12 | 0.07 | **0.03** | 0.62 | 0.04 | 0.07 | 0.04 | **0.04** | 0.56 |
| Mg | **0.03** | 0.10 | 0.05 | 0.04 | 0.21 | 0.03 | 0.06 | **0.03** | 0.05 | 0.18 |
| D best | | | 8/11 | | | | | 4/11 | | |

a difference of the size of the electronic heat capacity ($\gamma_e T \approx 0.4$–$1.2$ J mol$^{-1}$K$^{-1}$), which is not included. The self-consistent model gives the same $C_P$ and $B_T$ at 300 K; above 400 K its $C_P$ is lower because its $\alpha$ is (main text, Sec. V K).


[1] G. Raunio, L. Almqvist, and R. Stedman, "Phonon dispersion relations in NaCl," Phys. Rev. **178**, 1496–1501 (1969).

[2] National Bureau of Standards, "Molecular dynamics and structure of solids," NBS Special Publication 301 (National Bureau of Standards, Washington, DC, 1969).

[3] M. Kresch, M. Lucas, O. Delaire, J. Y. Y. Lin, and B. Fultz, "Phonons in aluminum at high temperatures studied by inelastic neutron scattering," Phys. Rev. B **77**, 024301 (2008).

[4] M. G. Kresch, *Temperature Dependence of Phonons in Elemental Cubic Metals Studied by Inelastic Scattering of Neutrons and X-Rays*, Ph.D. thesis, California Institute of Technology (2009).

[5] M. W. Chase, Jr., *NIST-JANAF Thermochemical Tables*, 4th ed., J. Phys. Chem. Ref. Data, Monograph 9 (1998) pp. 1–1951.

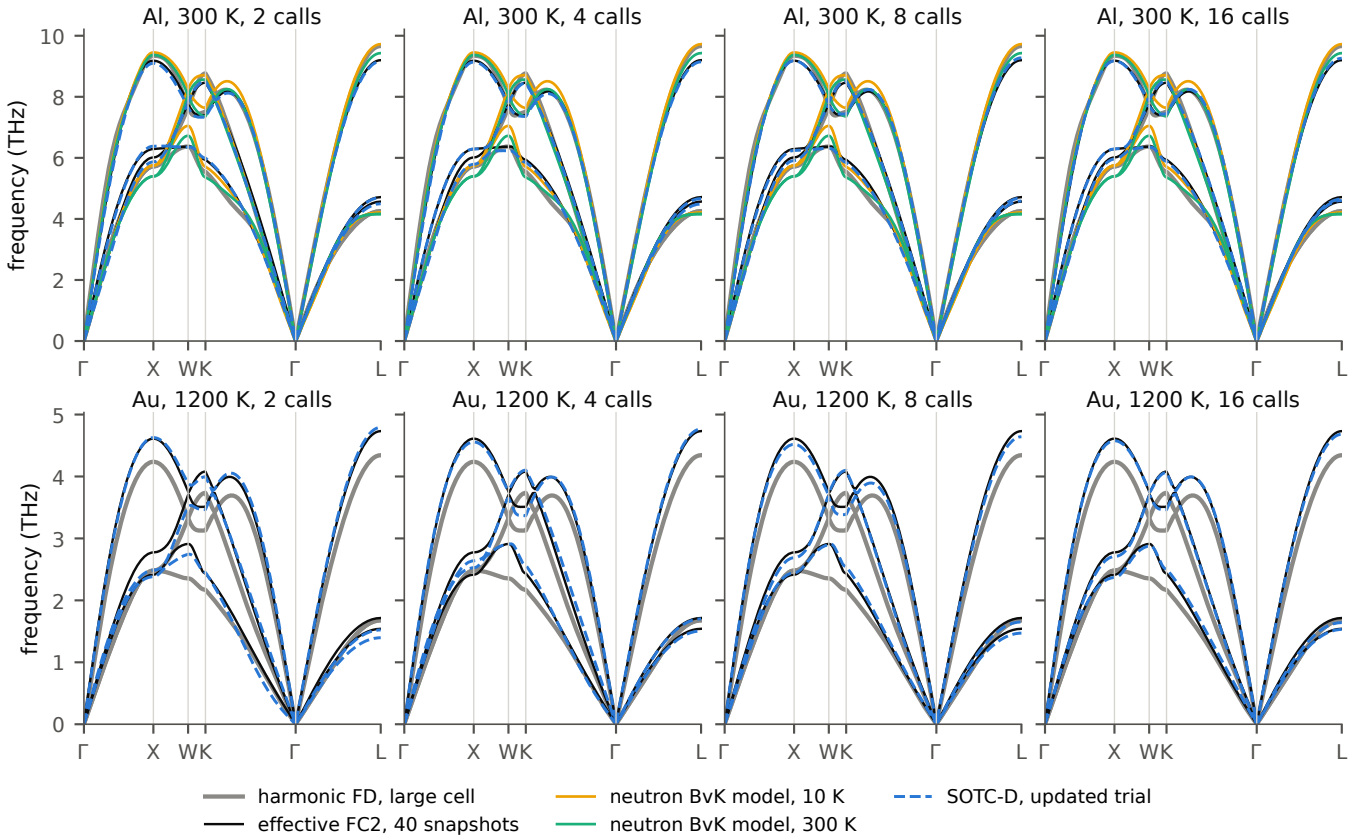



FIG. S4. Dispersions of SOTC-D from the updated trial (blue dashed) with 2, 4, 8, and 16 force calls, the converged large-cell harmonic FD reference (gray), the 40-snapshot iteration-1 effective FC2 (black), and, for Al, the Born–von Kármán models fitted to inelastic neutron data at 10 K (yellow) and 300 K (green). Same-cell models are Fourier interpolated from the 32-atom SOTC cell.

TABLE S8. Updated trial (iteration 1, crystal basis). Mesh RMSE (THz) against the iteration-1 effective FC2. Matched trial: SOTC-D designed from the iteration-1 trial ($D_1$) versus the better of random canonical subsets and pairs drawn from the same trial ($C_1$), scored against the disjoint effective FC2. Equal cost: bootstrapped SOTC-D (B; first pair from the harmonic trial, later pairs from the model refitted to the first pair, no extra force calls) and one-shot SOTC-D ($D_0$) versus canonical pairs from the harmonic trial ($C_0$; a single sTDEP iteration), scored against the 40-snapshot iteration-1 effective FC2, which shares no configuration with them. B and $D_0$ coincide at 2 calls by construction. The best entry of each group is bold.

| | matched, 4 | | equal cost, 4 | | | matched, 16 | | equal cost, 16 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $D_1$ | $C_1$ | B | $D_0$ | $C_0$ | $D_1$ | $C_1$ | B | $D_0$ | $C_0$ |
| Al | 0.07 | **0.06** | **0.02** | 0.06 | 0.06 | **0.03** | 0.04 | **0.02** | 0.02 | 0.03 |
| Ag | **0.04** | 0.08 | 0.08 | **0.06** | 0.08 | 0.05 | **0.05** | **0.04** | 0.06 | 0.10 |
| Au | **0.03** | 0.07 | **0.07** | 0.16 | 0.19 | **0.04** | 0.05 | **0.01** | 0.17 | 0.21 |
| Cu | **0.04** | 0.14 | 0.24 | 0.14 | **0.08** | **0.03** | 0.06 | 0.16 | **0.10** | 0.14 |
| Mo | **0.02** | 0.04 | 0.02 | **0.02** | 0.06 | **0.02** | 0.03 | **0.02** | 0.03 | 0.05 |
| W | **0.03** | 0.03 | 0.03 | **0.03** | 0.03 | **0.02** | 0.02 | 0.02 | **0.01** | 0.02 |
| NaCl | 0.09 | **0.08** | **0.07** | 0.21 | 0.15 | 0.07 | **0.06** | **0.04** | 0.14 | 0.12 |
| MgO | **0.13** | 0.18 | **0.27** | 0.37 | 0.31 | **0.10** | 0.12 | **0.10** | 0.32 | 0.25 |
| PbTe | **0.02** | 0.05 | **0.07** | 0.17 | 0.10 | **0.03** | 0.03 | **0.04** | 0.11 | 0.08 |
| Si | **0.05** | 0.07 | **0.06** | 0.09 | 0.09 | 0.06 | **0.04** | **0.04** | 0.09 | 0.06 |
| Mg | **0.04** | 0.05 | 0.03 | **0.03** | 0.04 | 0.04 | **0.04** | **0.02** | 0.04 | 0.03 |
| $D_1 < C_1$ / $B < C_0$ | 9/11 | | 9/11 | | | 7/11 | | 10/11 | | |
| held-out force $D_1 < C_1$ | 10/11 | | | | | 6/11 | | | | |

TABLE S9. Updated trial (iteration 1, supercell basis). Mesh RMSE (THz) against the iteration-1 effective FC2. Matched trial: SOTC-D designed from the iteration-1 trial ($D_1$) versus the better of random canonical subsets and pairs drawn from the same trial ($C_1$), scored against the disjoint effective FC2. Equal cost: bootstrapped SOTC-D (B; first pair from the harmonic trial, later pairs from the model refitted to the first pair, no extra force calls) and one-shot SOTC-D ($D_0$) versus canonical pairs from the harmonic trial ($C_0$; a single sTDEP iteration), scored against the 40-snapshot iteration-1 effective FC2, which shares no configuration with them. B and $D_0$ coincide at 2 calls by construction. The best entry of each group is bold.

| | matched, 4 | | equal cost, 4 | | | matched, 16 | | equal cost, 16 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $D_1$ | $C_1$ | B | $D_0$ | $C_0$ | $D_1$ | $C_1$ | B | $D_0$ | $C_0$ |
| Al | **0.10** | 0.21 | **0.07** | 0.09 | 0.15 | **0.08** | 0.10 | **0.06** | 0.06 | 0.10 |
| Ag | **0.15** | 0.15 | 0.12 | **0.12** | 0.17 | 0.10 | **0.10** | **0.08** | 0.09 | 0.13 |
| Au | **0.07** | 0.15 | **0.14** | 0.20 | 0.24 | **0.07** | 0.09 | **0.06** | 0.19 | 0.24 |
| Cu | **0.48** | 0.74 | 0.55 | **0.37** | 1.18 | **0.21** | 0.21 | 0.25 | **0.21** | 0.29 |
| Mo | **0.07** | 0.10 | 0.06 | **0.06** | 0.10 | **0.06** | 0.06 | 0.05 | **0.05** | 0.07 |
| W | **0.05** | 0.08 | **0.05** | 0.06 | 0.07 | **0.04** | 0.04 | 0.03 | **0.03** | 0.04 |
| NaCl | **0.14** | 0.19 | **0.16** | 0.30 | 0.23 | **0.10** | 0.11 | **0.07** | 0.16 | 0.15 |
| MgO | **0.23** | 0.38 | **0.33** | 0.44 | 0.44 | **0.21** | 0.22 | **0.16** | 0.36 | 0.30 |
| PbTe | **0.05** | 0.11 | **0.09** | 0.18 | 0.14 | **0.05** | 0.06 | **0.06** | 0.11 | 0.10 |
| Si | **0.05** | 0.07 | **0.06** | 0.09 | 0.09 | 0.06 | **0.04** | **0.04** | 0.09 | 0.06 |
| Mg | **0.04** | 0.05 | 0.03 | **0.03** | 0.04 | 0.04 | **0.04** | **0.02** | 0.04 | 0.03 |
| $D_1 < C_1$ / $B < C_0$ | 11/11 | | | 11/11 | | | 8/11 | | 11/11 | |
| held-out force $D_1 < C_1$ | 10/11 | | | | | | 6/11 | | | |

TABLE S10. Nearest-neighbor force constants (N/m): first- and second-shell longitudinal ($K_1$, $K_2$) and mean first-shell transverse ($K_{T1}$), averaged over all pairs of the shell. Experiment: Born–von Kármán fits to inelastic neutron data compiled in Ref. 4 (Cu interpolated linearly between 973 and 1336 K). All models at the production lattice parameters (300 K for Al, 1200 K for Cu).

| | Al, 300 K | | | Cu, 1200 K | | |
|---|---|---|---|---|---|---|
| | $K_1$ | $K_2$ | $K_{T1}$ | $K_1$ | $K_2$ | $K_{T1}$ |
| harmonic, large cell | 20.2 | 2.14 | -1.35 | 23.5 | 0.79 | -1.04 |
| harmonic, SOTC cell | 17.6 | 0.49 | 0.19 | 23.5 | 0.40 | -1.03 |
| effective FC2 | 18.8 | 0.86 | -0.51 | 30.1 | 0.04 | -1.57 |
| SOTC-D (8, updated) | 18.8 | 0.69 | -0.57 | 29.6 | 0.99 | -1.19 |
| bootstrapped (8) | 19.0 | 0.71 | -0.61 | 33.4 | 1.14 | -2.11 |
| neutron BvK | 20.1 | 2.22 | -1.36 | 25.6 | 0.73 | -1.93 |

TABLE S11. Al thermodynamics on the volume grid (PBE). $\alpha$ ($10^{-6}$ $K^{-1}$) of the converged 125-atom QHA from free-energy minimization and from the Grüneisen formula $\alpha = \sum_k \gamma_k c_k / 3BV$, evaluated with the static $B_0$ and $V_0$ (shortcut) and with $B_T$ and $V(T)$ of the minimization (exact). $B_T$ (GPa) and $C_P = C_V + TV\beta^2 B_T$ (J $mol^{-1}K^{-1}$, phonon part only) of the converged QHA and of the self-consistent SOTC-D (32 calls, independent-seed grid). Experiment: NIST-JANAF $C_P$ (Shomate fit, 298–933 K, including the electronic contribution)[5].

| | | $\alpha$, converged QHA | | $B_T$ | | $C_P$ | | |
|---|---|---|---|---|---|---|---|---|
| $T$ (K) | min. | Grüneisen, exact | Grüneisen, shortcut | QHA | SC | QHA | SC | expt. |
| 100 | 12.9 | 12.9 | 11.9 | 75.0 | 75.3 | 13.24 | 13.48 | – |
| 300 | 24.1 | 24.1 | 20.9 | 69.6 | 69.5 | 24.14 | 24.01 | 24.25 |
| 400 | 26.2 | 26.3 | 21.7 | 66.6 | 68.9 | 25.56 | 25.21 | 25.78 |
| 500 | 28.2 | 28.2 | 22.1 | 63.3 | 67.0 | 26.61 | 25.87 | 26.84 |
| 700 | 32.9 | 32.9 | 22.4 | 55.8 | 74.0 | 28.62 | 26.96 | 29.10 |
| 900 | 40.2 | 40.1 | 22.6 | 46.6 | 58.5 | 31.34 | 27.02 | 32.31 |